Try it and explain your thinking

A fictional wear check shows 8.0 mm initially and 6.5 mm now

A fictional wear check shows 8.0 mm initially and 6.5 mm now. Find the loss. Does that establish replacement?

Skills you use: Numeracy, Problem solving.

Identify which maintenance items belong to the actual screed and which to shared tractor systems.

Identify which maintenance items belong to the actual screed and which to shared tractor systems.

Skills you use: Reading, Writing.

Draft a safe cleaning handover between two operators.

Draft a safe cleaning handover between two operators.

Skills you use: Communication, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 3 task criteria out of 6; require at least 5 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Service actual lubrication points and shared systems safely.	0 / 1 / 2	Evidence/help: _____
Inspect plate condition against manufacturer criteria.	0 / 1 / 2	Evidence/help: _____
Clean, restore guards and complete a service/defect record.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, J4, printed p. 86; source PDF pp. 88–88.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline J5

Operate asphalt screed

Screed operation combines careful setup with measured feedback. The learner practises on the selected machine under supervision, using its OMM for exact control actions and limits.

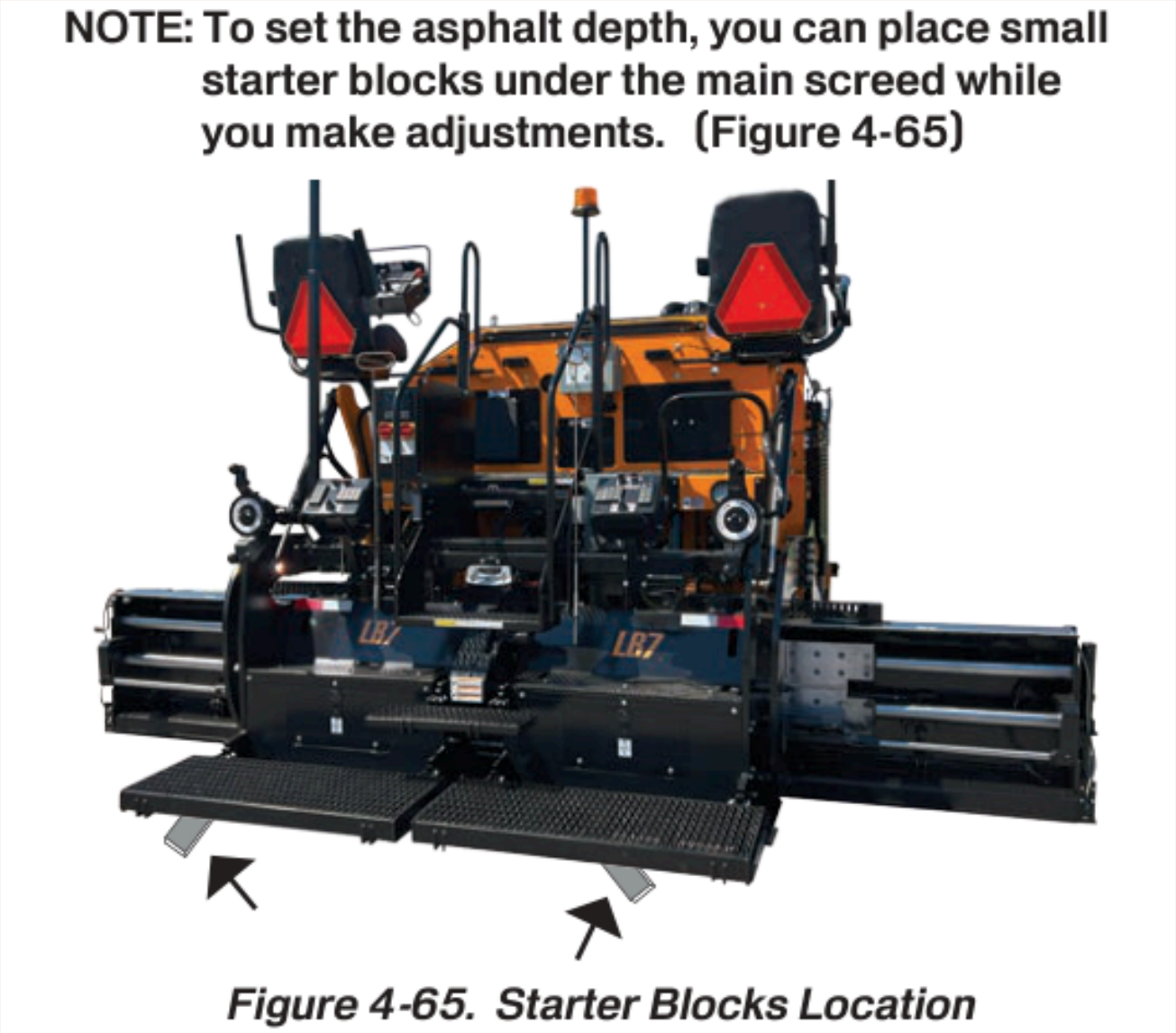
- Set-up screed
- Maintain proper thickness
- Check depth of mat

LEARNING STEP 1

Set-up screed

Inspect the screed and surroundings, identify the fitted heating system and complete daily maintenance. Establish starting width, crown/contour, extension alignment and the trial-based loose-thickness setting. Lower onto the specified starting supports through the safe procedure. Operate heating only as instructed; protect against burns, fire and electrical hazards. The LeeBoy 8680 example identifies separate main/extension/end-gate heat controls and warns against overheating. Do not transfer its settings to another machine.

LeeBoy 8680 starter blocks and screed locks



Note. LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP, p. 4-57; PDF p. 125). Serial 365152 and above. Faithful crop; use the complete matching procedure. Full source context.

LEARNING STEP 2

Maintain proper thickness

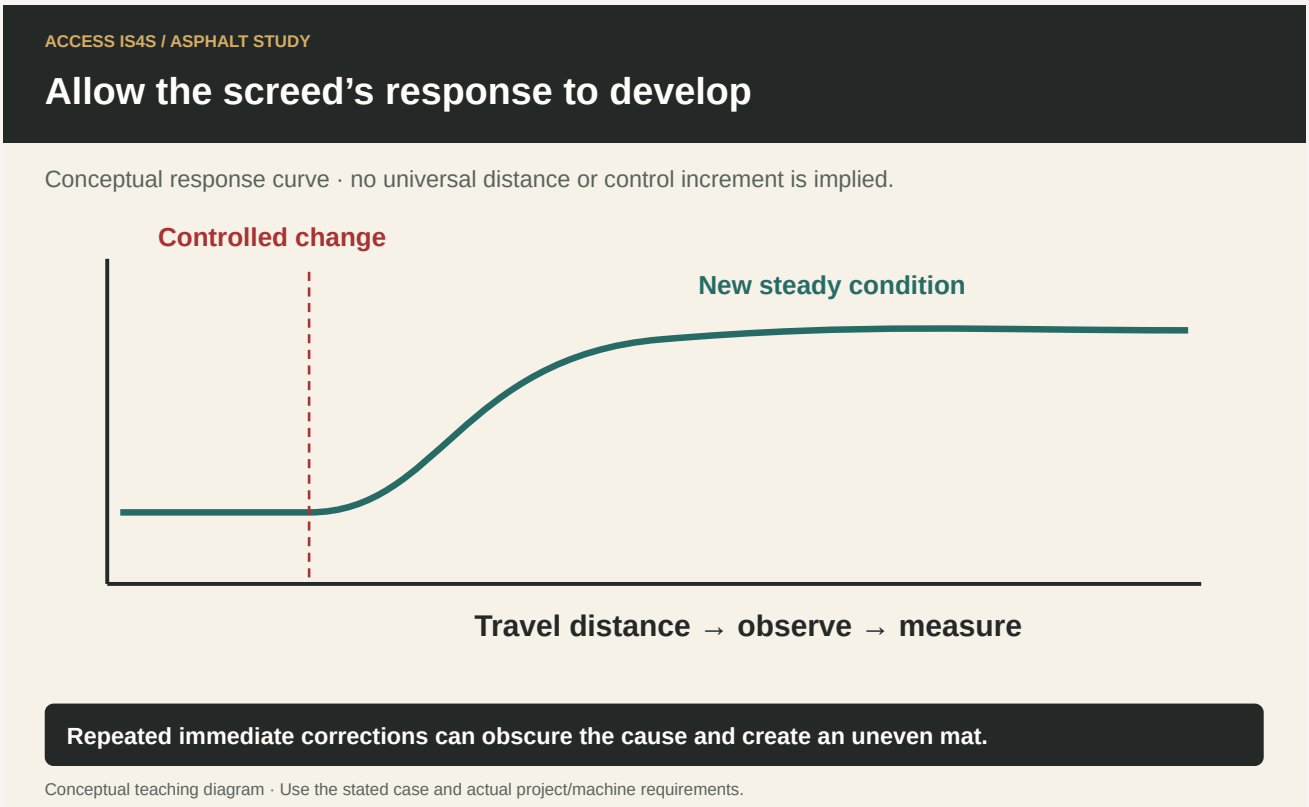
Coordinate with the paver operator to maintain a suitable steady speed and material head. Monitor the mat, hydraulic/electrical indications and assigned controls. Make authorized small, deliberate changes to angle of attack, tow points or contour as appropriate, then allow the response to develop and recheck. Keep width changes coordinated with auger/feed distribution and safe clearance. Operate lights or other ancillary equipment only where fitted and included in the plan.

LEARNING STEP 3

Check depth of mat

Measure loose depth using the approved probe/tool and safe access method. Identify the base interface rather than pushing into soft support, record station/side and avoid contact with hot material or moving equipment. A probe reading is not a core result or proof of final compacted thickness. Compare repeated readings with the specified target and trial relationship, report departures and retain the record.

Allow the screed’s response to develop



Note. ACCESS IS4S. (2026). Allow the screed’s response to develop [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Loose depth and final thickness

A verified trial relationship is loose/compacted = 1.25. Target compacted thickness is 40 mm. A loose-depth check reads 50 mm.

Think it through.

Estimated loose target = 40 × 1.25 = 50 mm, so the reading matches that exercise target. Final compacted thickness still requires the specified verification; one loose reading does not establish the whole mat’s acceptance.

Try it and explain your thinking

For the same fictional factor, estimate loose thickness for a 48 mm compacted target.

For the same fictional factor, estimate loose thickness for a 48 mm compacted target.

Skills you use: Numeracy, Reading.

Explain why repeated immediate adjustments after each probe reading can be unhelpful.

Explain why repeated immediate adjustments after each probe reading can be unhelpful.

Skills you use: Problem solving, Communication.

Record a supervised setup and two subsequent checks in a digital table.

Record a supervised setup and two subsequent checks in a digital table.

Skills you use: Digital, Writing, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Precheck and service the screed; operate its fitted heater safely.	0 / 1 / 2	Evidence/help: _____
Set thickness/crown/width and starting supports under supervision.	0 / 1 / 2	Evidence/help: _____
Coordinate speed and material head; monitor hydraulics and fitted ancillary controls.	0 / 1 / 2	Evidence/help: _____
Measure loose depth safely and evaluate an authorized adjustment.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, J5, printed p. 87; source PDF pp. 89–89.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- ThermoWorks. (n.d.). Industrial IR with Circle Laser: IR-IND / IRK-2 Operating Instructions. Open publication.
- BC Ministry of Transportation and Transit. (2025). 2025 Standard Specifications for Highway Construction, Volume 1. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

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Study chapter · Outline J6

Follow shut down procedures

Screed shutdown and cleaning are coordinated with the paver. The final state must protect people from heat, movement and unsupported components while keeping the equipment ready for inspection.

Shut down procedures

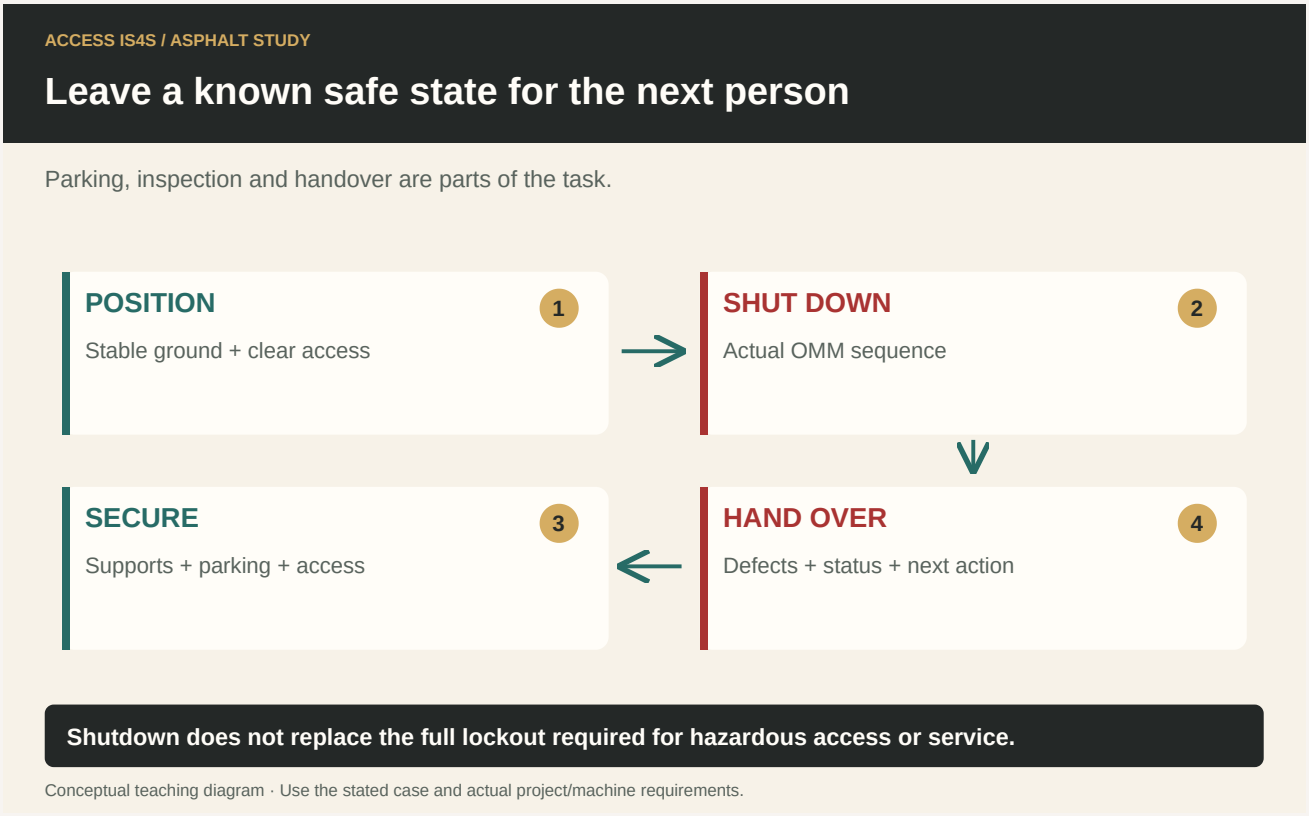
Clean screed after use

LEARNING STEP 1

Shut down procedures

Choose the approved parking position with service access and suitable ground. Follow the actual stopping, heater shutdown, component positioning/support and paver parking sequence. Inspect safely for leaks, damage and missing guards. Apply full lockout when the cleaning or service task requires it; normal shutdown alone is insufficient. Secure controls and access and record the final state.

Leave a known safe state for the next person

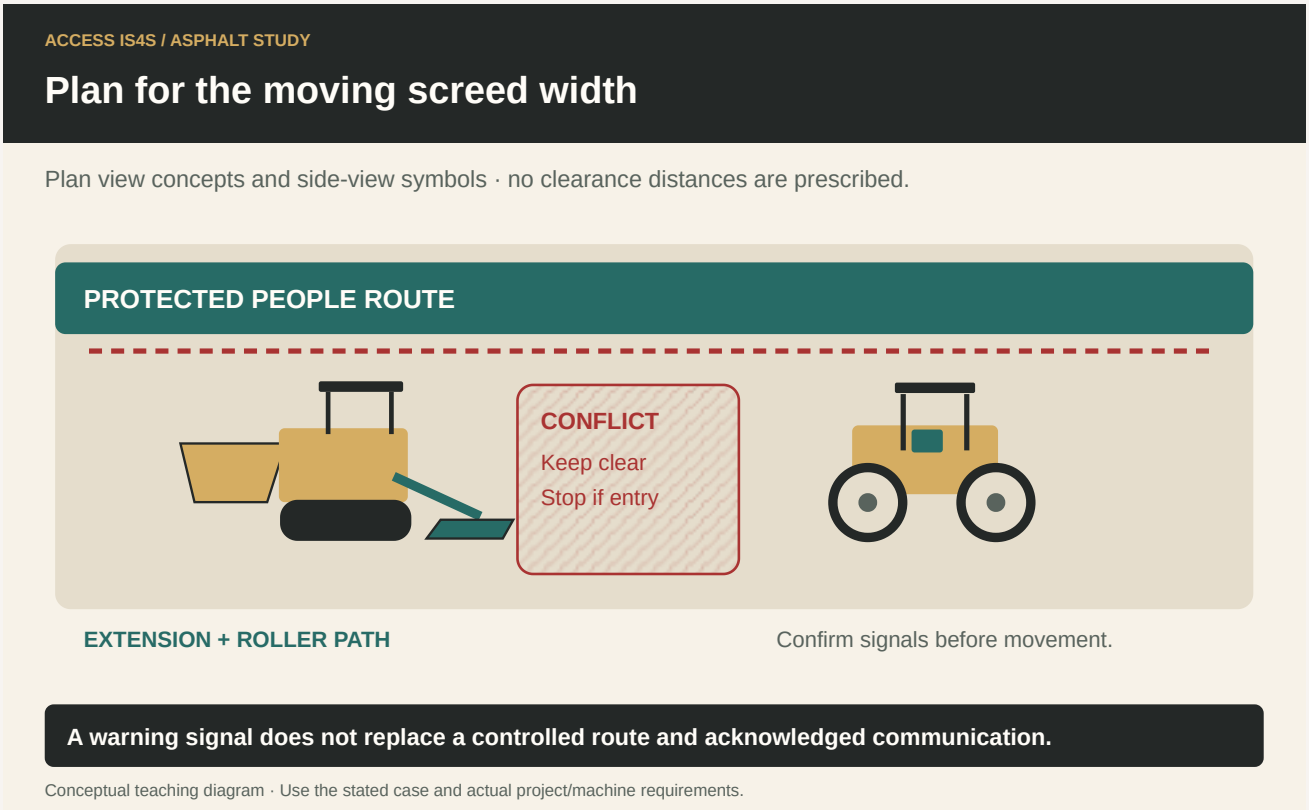


Note. ACCESS IS4S. (2026). Leave a known safe state for the next person [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Clean screed after use

Clean buildup using the manufacturer-approved method, compatible product and SDS controls. Coordinate with the paver crew so shared systems cannot move unexpectedly. Control ignition sources and residual heat; never spray a flammable product onto a hot heater or open flame. A fixed “20-minute clean” is not a universal safe duration. Collect scrapings and wash/cleaning waste, protect drainage and restore guards. Record defects and any work still required before the next start.

Plan for the moving screed width



Note. ACCESS IS4S. (2026). Plan for the moving screed width [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · Cleaning is delayed

The screed remains too hot for the approved cleaning product and procedure when the scheduled shift ends.

Think it through.

Maintain the safe shutdown state and coordinate the required cooling and cleaning arrangements. Do not spray early to meet the schedule. Record the outstanding task and responsible person so the next shift does not assume cleaning is complete.

Try it and explain your thinking

Explain three checks before cleaning starts.

Explain three checks before cleaning starts.

Skills you use: Reading, Problem solving.

An assessment has 20 possible points and earns 15

An assessment has 20 possible points and earns 15. What percentage is that, and what follows?

Skills you use: Numeracy.

Write a handover stating that cleaning remains incomplete.

Write a handover stating that cleaning remains incomplete.

Skills you use: Writing, Communication, Collaboration.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

The outline specifies a minimum 80% checklist result for this task. Score the 3 task criteria out of 6; require at least 5 points and all critical items passed. This local scoring rubric operationalizes the checklist; it is not a separate certification. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Complete the matching shutdown and support procedure.	0 / 1 / 2	Evidence/help: _____
Coordinate safe cleaning with heat/energy and waste controls.	0 / 1 / 2	Evidence/help: _____
Secure the equipment and record incomplete work.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, J6, printed p. 88; source PDF pp. 90–90.
- Caterpillar. (Various editions). Paving, compaction, milling and machine publications; identify the applicable document/model. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 10: De-energization and Lockout. Open publication.
- LeeBoy. (2026). 8680 Commercial Paver: Safety and operations manual (1033428-SOP). Serial 365152 and above. Components pp. 3-1–3-32; operation pp. 4-1–4-68; maintenance section 5. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline K1

Raker roles and responsibilities

The handwork crew shapes and finishes areas the paver cannot complete on its own. A raker needs judgement about material, grade, joints and timing, as well as awareness of every machine moving nearby.

Typical raker duties

LEARNING STEP 1

Typical raker duties

Typical duties include preparing small areas, distributing mix around structures, finishing edges, correcting authorized local irregularities and preparing longitudinal or transverse joints. A lute commonly has a long handle and a broad head with straight and toothed edges; a rake, shovel, tamper and wheelbarrow serve different purposes. Inspect handle attachment, head/brace condition and working edges before use.

Work around manholes and other appurtenances to the approved elevation and drainage detail. Remove excess and square the end of a run through the planned method. Avoid scattering coarse material over a finished mat or repeatedly disturbing a surface until the mixture cools. A paver or screed problem that repeats needs communication and correction at its cause, not endless hand repair.

At joints, understand which face is prepared, which treatment is specified, what overlap/allowance is intended and how compaction will occur. Longitudinal joints run generally along the paving direction; transverse joints cross it. Their details and rolling sequence depend on the project and mixture. Do not infer a universal joint allowance from a sketch.

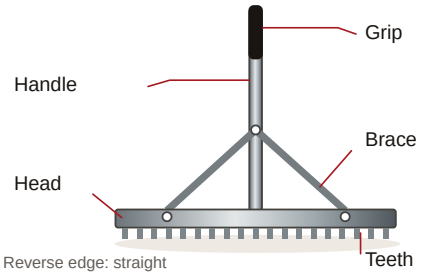
Coordinate with the paver, screed and roller operators before entering an area. Keep tools and handles out of traffic and machine paths. Preserve accessible routes and protect nearby drains. Careful handwork serves the finished road and the people using it; communicate a concern early rather than hiding it under a cosmetic finish.

Name the parts of a handwork tool

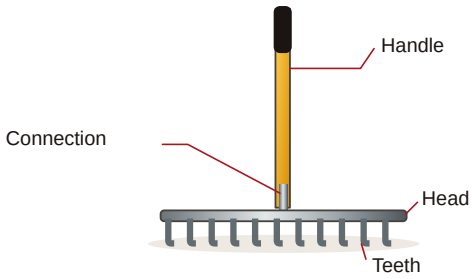
ACCESS IS4S · ASPHALT HANDWORK

The small tools that shape the work

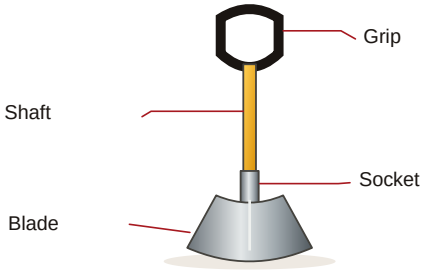
01 / LUTE



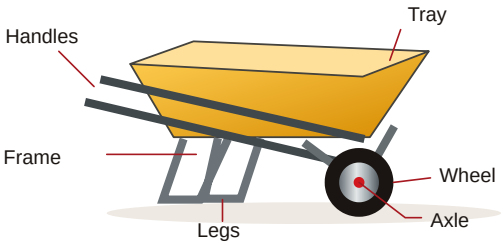
02 / RAKE



03 / SHOVEL



04 / WHEELBARROW



ILLUSTRATIVE PART NAMES · Verify the actual model, parts list and condition before use.

Note. ACCESS IS4S. (2026). Handwork tool anatomy [Original instructional SVG]. Based on the Seymour Midwest tool information retained in the tool library. Full source context.

Rakes And Handwork



Note. Seymour Midwest. (2026 publisher download). Partner Resource Guide (source PDF p. 156). https://seymourmidwest.com/uploads/SMW2026_PRGweb.pdf Faithful source crop. Full source context.

Worked example · The same hollow returns

A small low area appears repeatedly behind the same screed location. The raker has filled it twice.

Think it through.

Report the repeated location and timing to the screed/quality lead. Complete only authorized hand correction while the underlying cause is investigated. Repeated filling without diagnosis can change texture and compaction and hide a continuing setup issue.

Try it and explain your thinking

Choose a lute, shovel, tamper and wheelbarrow for four distinct purposes.

Choose a lute, shovel, tamper and wheelbarrow for four distinct purposes.

Skills you use: Reading, Problem solving.

A rectangular small area is 2.4 m × 1.5 m

A rectangular small area is 2.4 m × 1.5 m. Find area for the work plan.

Skills you use: Numeracy.

Prepare a photographed or SVG tool-identification card with AI assistance if available.

Prepare a photographed or SVG tool-identification card with AI assistance if available.

Skills you use: Digital, Writing, Creativity and innovation.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Identify and inspect handwork tools.	0 / 1 / 2	Evidence/help: _____
Explain local finishing, structure and joint duties.	0 / 1 / 2	Evidence/help: _____
Report a repeated defect and coordinate safe access.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, K1, printed p. 89; source PDF pp. 91–91.
- Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4), Chapter 8: Hot Mix Asphalt Paving; source PDF pp. 391–446. Open publication.
- Caterpillar. (2016). Guide to Asphalt Paving (QEBQ1888). Asphalt paving process, tractor, screed, material delivery and handwork; model examples vary. Open publication.
- Seymour Midwest. (2026 publisher download). Partner Resource Guide. Open publication.

Figure-specific pages, editions and original contributors are credited beside each image. Current project requirements and the matching machine instructions determine practical use.

Study chapter · Outline K2

Raking safety procedures

Handwork places a person close to hot material and moving equipment while using long tools. Plan the work position and escape route before concentrating on the surface.

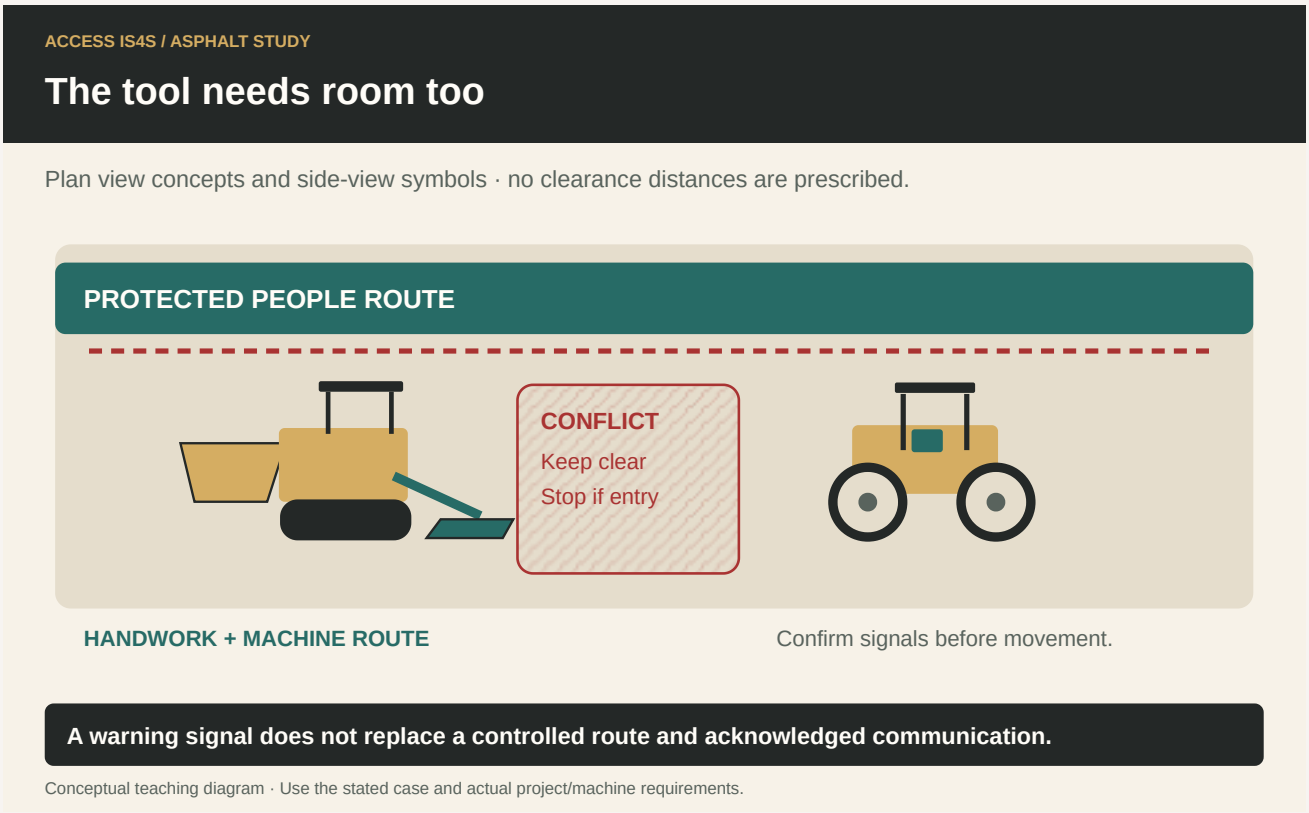
- Interpret job planning
- Raker-specific safety hazards
- Traffic control requirements
- Proper PPE

LEARNING STEP 1

Interpret job planning

Read the path, work pattern, thickness, temperature requirements, crown, obstacles and sequence. Agree when handwork can enter and when the roller needs the area. Identify the safe material-transfer route, including wheelbarrow movement. Protect public access and confirm how a changing condition will be communicated.

The tool needs room too



Note. ACCESS IS4S. (2026). The tool needs room too [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

LEARNING STEP 2

Raker-specific safety hazards

Recognize burns, heat strain, fumes, noise, dust, pressurized systems, blind areas, pinch zones and traffic. Keep long handles inside the work area. Use manageable loads and a stable stance, move feet to turn rather than twisting under load, and vary tasks/breaks through the work plan. Do not overreach to save a step. Wet or unsuitable footwear can worsen hot-material exposure. Report fatigue, dizziness or damaged equipment early.

LEARNING STEP 3

Traffic control requirements

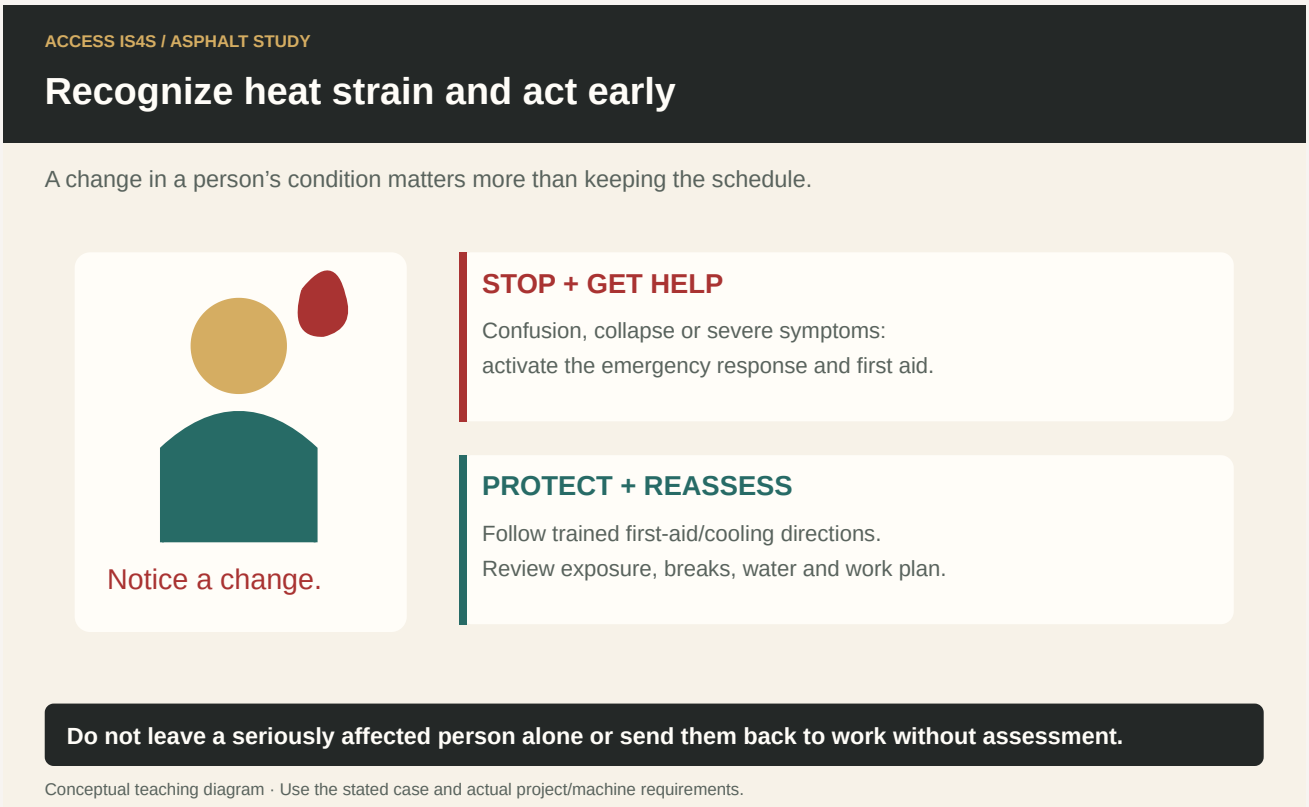
Coordinate with trained traffic-control personnel and the designated equipment signallers. A raker does not step into live traffic to retrieve a tool or direct a truck without the assigned role and controls. Obtain acknowledgement before approaching a machine’s working area. If visibility or communication is lost, stop the affected activity and move only through the established safe route.

LEARNING STEP 4

Proper PPE

Use task-selected footwear, head, eye, hearing, visibility and heat/chemical protection. Inspect condition, fit and contamination. Follow current Part 8 supply responsibilities and employer arrangements. PPE does not replace ergonomic planning or separation from equipment. The instructor observes actual safe tool handling, communication and movement, not only whether the learner can name PPE.

Recognize heat strain and act early



Note. ACCESS IS4S. (2026). Recognize heat strain and act early [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Worked example · A handle reaches into the lane

A worker turns with a long lute while watching the mat. Its handle crosses the work-zone boundary.

Think it through.

Stop and reposition through the safe route. Review the work space, tool orientation and traffic boundary with the lead. A safe stance must account for the whole tool, not only the worker's feet.

Try it and explain your thinking

Identify three ergonomic improvements for shovelling a small patch.

Identify three ergonomic improvements for shovelling a small patch.

Skills you use: Problem solving, Reading.

A practice plan has 40 min of handwork split into four equal periods

A practice plan has 40 min of handwork split into four equal periods. How long is each period? Does that prescribe a heat-rest schedule?

Skills you use: Numeracy.

Rehearse a request for safe access to finish an edge before rolling.

Rehearse a request for safe access to finish an edge before rolling.

Skills you use: Communication, Collaboration, Adaptability.

Show the skill in supervised practice

Machine/model/serial or classroom case: _____ · Manual/procedure and revision: _____ · Date/location: _____ · Learner/observer: _____

Record each criterion as 0—not yet demonstrated, 1—with task-solving support, or 2—independently under the stated conditions. Mark unobserved criteria separately; do not convert missing opportunity into a zero. The assessor determines the required repeat practice and applicable qualification requirements. A critical failure means hold, correction and reassessment regardless of the numerical result. Use safe simulations for classroom reasoning; actual operation requires the selected equipment, approved conditions and competent supervision.

What the observer checks	Result	Evidence
Critical: Safe access, surroundings and communication maintained.	Pass / hold	Observation: _____

What the observer checks	Result	Evidence
Critical: Required energy, heat, movement and public-access controls maintained for the actual task.	Pass / hold	Observation: _____
Use correct PPE and demonstrate ergonomic tool handling.	0 / 1 / 2	Evidence/help: _____
Coordinate with traffic control and equipment operators.	0 / 1 / 2	Evidence/help: _____
Keep people and long tools within the controlled work area.	0 / 1 / 2	Evidence/help: _____

Actual conditions and support: _____

Feedback, next practice and authorized release/status: _____

Sources and further reading

- SkilledTradesBC. (2016). *Asphalt Paving/Laydown Technician program outline*, K2, printed p. 90; source PDF pp. 92–93.
- Canadian Centre for Occupational Health and Safety. (retrieved October 30, 2026). Hand Tools: General Hand Tool Operation. Open publication.
- Canadian Centre for Occupational Health and Safety. (retrieved October 30, 2026). Shovelling. Open publication.
- WorkSafeBC. (2026; R05/26). Preventing Heat Stress at Work. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 8: Personal Protective Clothing and Equipment. Open publication.
- WorkSafeBC. (Web snapshot: October 30, 2026). OHS Regulation, Part 16: Mobile Equipment. Open publication.

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Study chapter · Outline K3

Place asphalt hot mix

Hand placement aims for a consistent layer that can be compacted properly. Work promptly and deliberately, using the approved preparation, joint and compaction methods for the actual repair or small area.

Common hand work crew duties

Common techniques

LEARNING STEP 1

Common hand work crew duties

Prepare the road edge, driveway, intersection, trench patch or pothole according to the specified repair detail. Confirm sound boundaries, support, drainage, depth and the required bond treatment. Apply the specified tack/emulsion to prepared faces under its procedure; an inert simulation may be used in training. Use shovels to place manageable amounts close to their final position. Coordinate delivery without entering a truck’s uncontrolled discharge area.

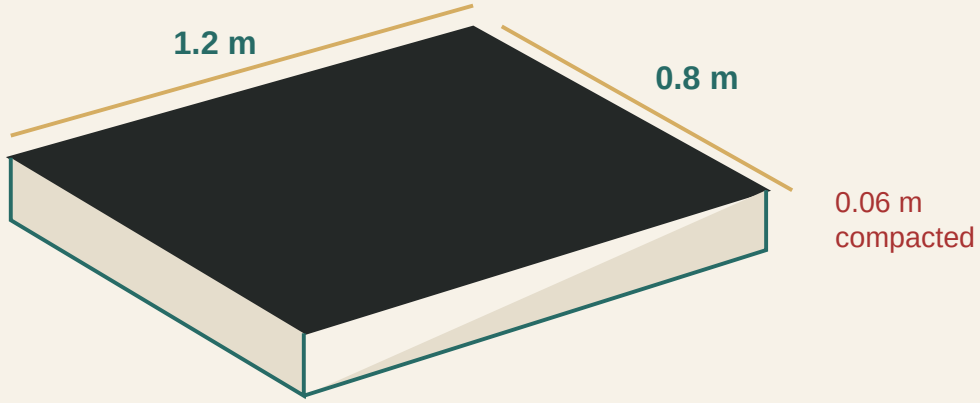
A wheelbarrow can move small quantities where the route, load, footing and heat controls allow. Inspect wheel, handles and tray, keep the load manageable and use the designated transfer arrangement. Spread with the suitable toothed or straight lute edge for the material and task. Avoid excessive manipulation that separates particles or cools the mix. Grade changes need controlled placement to limit uneven compaction, not an arbitrary heap of loose material.

Plan the patch in three dimensions

ACCESS IS4S / ASPHALT STUDY

Plan the patch in three dimensions

Fictional rectangle: 1.2 m × 0.8 m × 0.06 m compacted depth.



Volume = 1.2 × 0.8 × 0.06 = 0.0576 m³

This is geometry. The repair detail determines materials, lift method and compaction.

Conceptual teaching diagram · Use the stated case and actual project/machine requirements.

Note. ACCESS IS4S. (2026). Plan the patch in three dimensions [Original instructional SVG]. Conceptual explanation based on the cited chapter readings; geometry and operating arrangements are illustrative, not prescribed settings.

Rakes And Handwork



Note. Seymour Midwest. (2026 publisher download). Partner Resource Guide (source PDF p. 156). https://seymourmidwest.com/uploads/SMW2026_PRGweb.pdf Faithful source crop. Full source context.

LEARNING STEP 2

Common techniques

“Bumping” in this teaching context means controlled local adjustment of the loose mat with the approved hand tool to bring material into the required line or shape. The instructor demonstrates the project’s technique and terminology; it is not permission to strike machinery or people. Ramping forms a planned transition, which may be temporary or permanent according to the detail. Work around manholes to the verified elevation and crossfall without trapping water.

Square and prepare the end of a run and joints using the accepted method. Coordinate tamping/compaction equipment and roller access while the mixture remains suitable. A tiger torch or other heating tool is used only where specifically authorized, with its own training, equipment instructions and

fire controls. It is not a default way to restore cooled or unsuitable mix. Inspect the finished line, surface and drainage, and report defects for authorized correction.

Tampers



Note. Seymour Midwest. (2026 publisher download). Partner Resource Guide (source PDF p. 91). https://seymourmidwest.com/uploads/SMW2026_PRGweb.pdf Faithful source crop. Full source context.

Worked example · A small rectangular patch

A prepared patch is 1.2 m × 0.8 m with 0.06 m compacted depth.

Think it through.

Area = 0.96 m²; compacted volume = 0.0576 m³. Material quantity requires an appropriate density and justified allowance. Confirm the repair detail and compaction method; the volume calculation does not establish that a single lift or any particular tool is suitable.